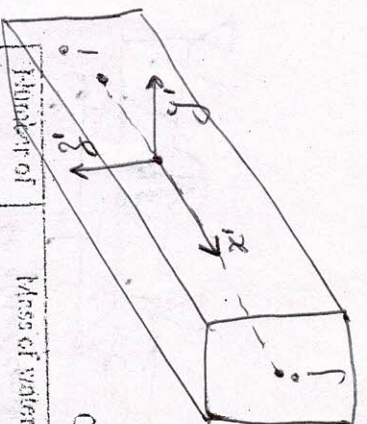


Number of observations	Mass of water collected			Time is sec.	
	120°	Hemi-spherical (180°)	Flat	120°	Hemi-spherical
1					
2					
3					
4					
5					
6					

CALCULATIONS:

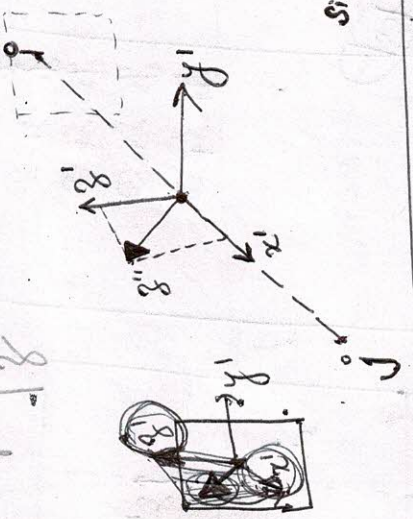
- (a) Actual discharge (Q_a) = v/t ; k = coefficient of vane
- (b) Velocity of flow (V) = Q_a/A ; a = area of vane
- (c) Theoretical force (F^t) = $k\rho a v^2$; v = velocity

$[d_{1m}]$, $[d_{2m}]$: ele. slope = ele. slope

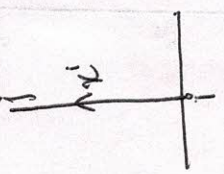
avoid δx local in unit provided vector of global axes that

parallel to this. $(T^0_0 T^0_0) = \delta$ $(T^0_0 T^0_0) = \delta$ $(T^0_0 T^0_0) = \delta$

local δ local δ $\delta \in h, h, x \in h$ $\delta \in x, x, h$ (δ) and δ, x is separable and δ



local	color
δ	δ
δ	δ
δ	δ



global vector to

is separable to

convert δ stream problem into reinforcement

Net the material between cone

Nonlinearity $\rightarrow$ Material $\rightarrow$ Geometric

2D and 3D

NITMA // IDA

Divide structures according to some level

use seismic parameters accordingly.

classification of structure to low high strength levels

build same structures with fiber based approach and lumped plasticity approach

main purpose (choose effective parameters) verify every iterations

low frequency modes diff. seismic parameters
high frequency

Damage Index : Envelope : Peak & Angle

ANN : CNN : from multiple NN, Train again & ~~retrain~~ ANN on the output of previous to get another output

Failure Modes

- (i) Soft Storey
- (ii) Strong Beam weak Column
- (iii) ~~Weak beam~~ ~~and~~ ~~the~~ ~~Part~~ ~~&~~ ~~Angle~~ : Artificial Damage
- (iv) shear Failure in columns
- (v) short Column effect.

Buckling is not particularly an issue in NPP

→ 18% diff

Collect IDA & NITMA buildings ~~and~~ analysed with operators.

Inter-Storey Drift ($\geq 2\%$)

Soft Storey failure.

very commonly repeated.

$$DI_{local} = \frac{\theta_m - \theta_r}{\theta_u - \theta_r} + \frac{\beta}{M_y \theta_u} E_T$$

θ_m = earthquake induced peak curvature
beyond load history.

θ_r = recoverable curvature / yield

θ_u = ultimate "

M_y = yield moment

E_T = dissipated hysteretic energy

θ_r, θ_u, M_y : Moment-curvature analysis.

β : 0.1-0.15 (Constant)

$$DI_{global} = \frac{\sum_{i=1}^n DI_{local} E_i}{\sum_{i=1}^n E_i}$$

$$\sum_{i=1}^n E_i$$

$$DI_{system} = \max. \left(\frac{V(t)}{V_{cap}} \right)^n$$

Therefore

Capacity exponent (2-3)

Capacity (ACI EC2 or Hines et al)

Can provide values for n for future researchers

N, mm, s

$$24 \text{ kN/m}^3$$

$$\frac{24 \times 10^3 \text{ N}}{10^9 \text{ mm}^3}$$

$$24 \times 10^{-6} \text{ N/mm}^3$$

$$24 \times 10^{-5} \text{ N/mm}^3$$

F = ma

$$N = kg \frac{m}{s^2}$$

$$\frac{N s^2}{mm} = \text{mass}$$

$$N = (kg) \frac{m}{s^2}$$

$$(kg) = \frac{N s^2}{m} = \left(\frac{N s^2}{1000 \text{ mm}} \right)$$

$$1kg = \frac{1}{1000} \frac{N s^2}{mm}$$

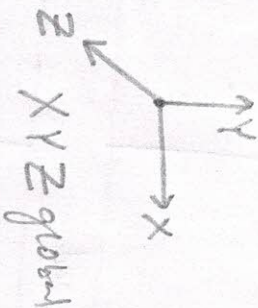
$$g = \frac{mm}{s^2}$$

$$g = \frac{9.8 \text{ m}}{s^2} = 9800 \left(\frac{mm}{s^2} \right)$$

$$1 \text{ MPa} = 1 \text{ N/mm}^2$$

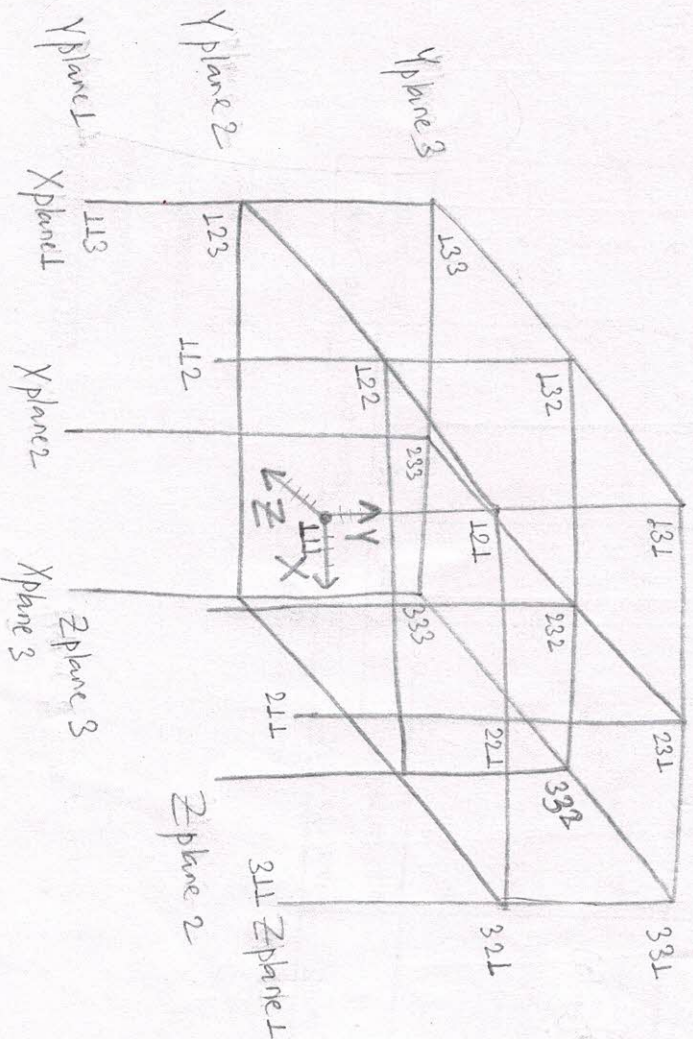
$$\text{Modulus (E, G)} = \text{N/mm}^2 = \text{MPa}$$

$$\text{Energy N} \cdot \text{mm}$$



Nomenclature

Nodes: (plane X) (plane Y) (plane Z)



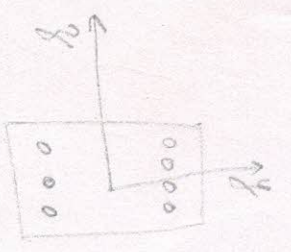
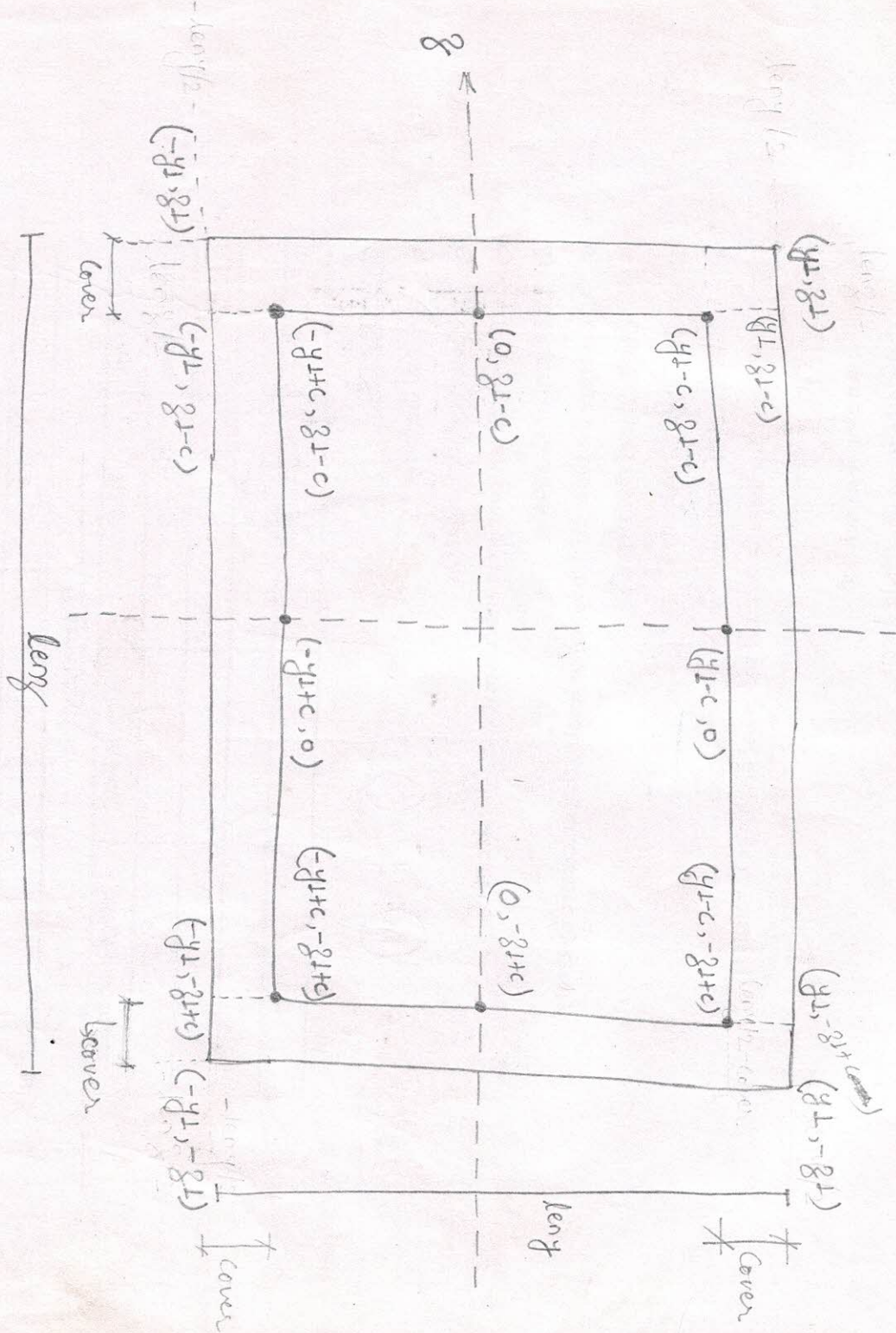
Nodes: (X plane) (Y plane) (Z plane)

Element: (Start X plane) (End X plane) (Start Y plane) (End Y plane) (Start Z plane) (End Z plane)

$$y_1 = \text{leny} / 2$$

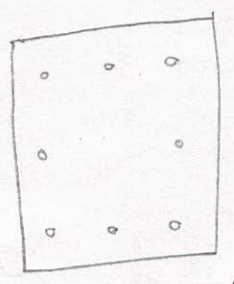
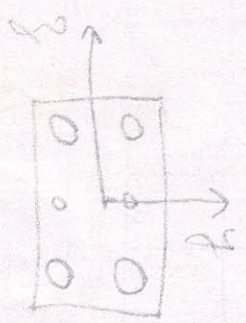
$$g_1 = \text{leny} / 2$$

$$\text{Cover} = c$$



$$g \rightarrow x$$

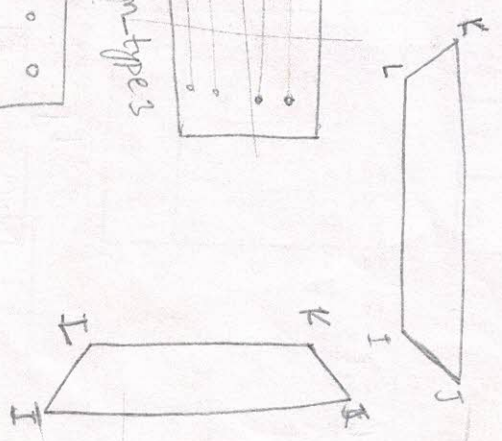
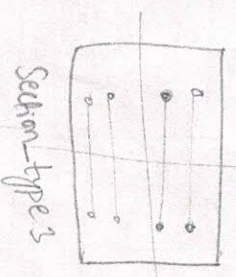
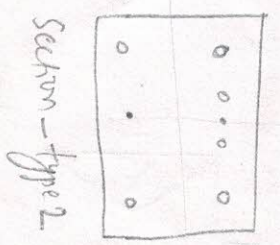
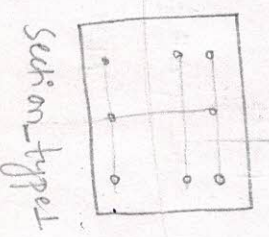
$$b \rightarrow y$$



$$\text{Col} \begin{matrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{matrix}$$

$$X_{\text{Beam}} \begin{matrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{matrix}$$

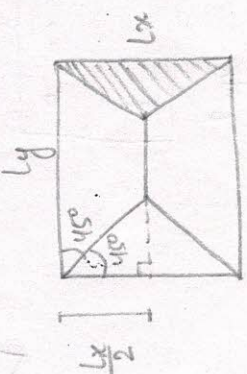
$$Y_{\text{Beam}} \begin{matrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{matrix}$$



$$\left(\frac{4}{x_7}\right) 10 = \frac{x_7}{7} * \frac{4}{x_7} * 10$$

$$\frac{4}{x_7} = \frac{2}{x_7} * x_7 * \frac{2}{7} = \nabla \text{ Area}$$

$$\frac{2}{x_7} = 4$$



Observation table

Number of observations	Head H mm	Mass of water collected kg.	Time
1			
2			
3			
4			
5			
6			

$$\frac{2}{x_7} * \frac{2}{x_7 - h_7 + h_7}$$

Observation table

Number of observations	X mm	0	5	10	15	20	25	30	35	40
1	Y mm									
2	Y mm									
3	Y mm									

Observation table

Number of observations	Head mm	Time in Sec.	Average time in Sec.
	H ₁	H ₂	
1			
2			
3			

CALCULATION:

- Actual discharge Q_a .
- Theoretical discharge Q_{th} .
- C_d for each test conducted and compute average value of C_d .
- Determine C_v and C_c .
- Compare the time taken to lower the level to that calculated by theory.
- Coefficient of discharge C_d .
- Coefficient of velocity C_v .
- Coefficient of contraction C_c .
- Plot actual discharge Q versus H and actual discharge Q versus $H^{1/2}$.

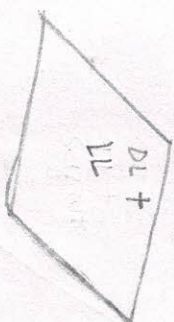
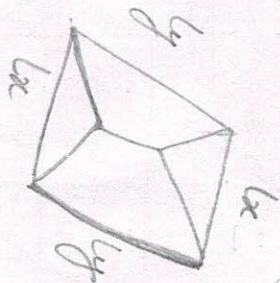
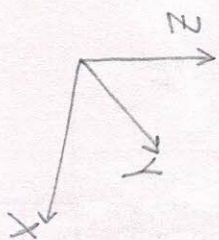
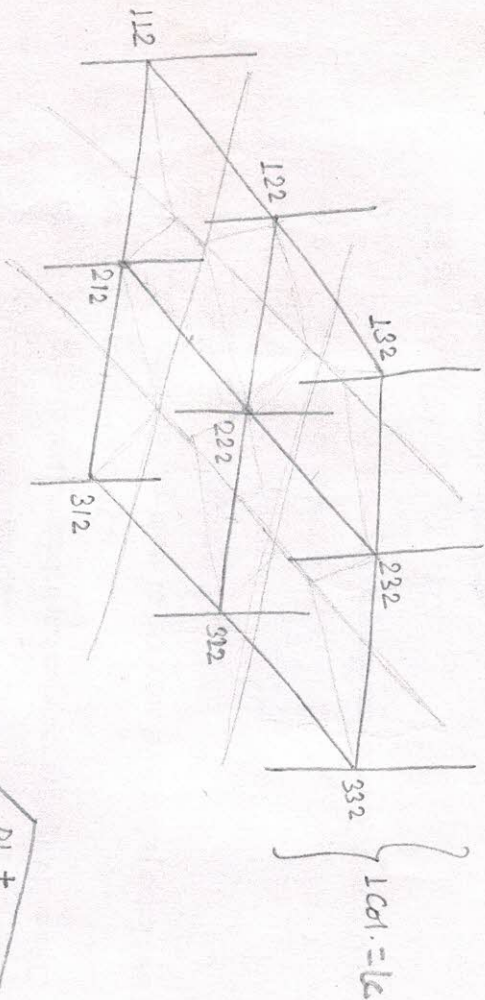
3 25 6 10 9 45 16

PRESENTATION:

- Show a sample calculation.
- Present the result in a tabular form.

$$\frac{1 \text{ kN}}{\text{m}^2} = \frac{1000 \text{ N}}{106 \text{ mm}^2}$$

$$1 \text{ kN/m}^2 = 10^{-3} \text{ N/mm}^2$$



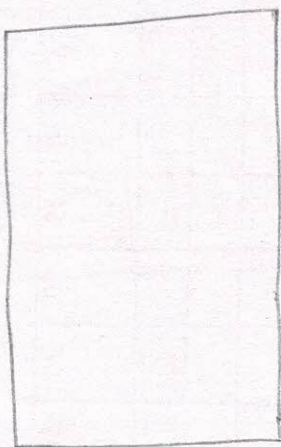
DL: Floor finish load +
furniture load (X)
+ self wt.

Self wt. = $\gamma_c \times \text{thickness}$
Floor finish = 1 kN/m^2

LL = 0 kN/m^2 for every floors
except top.
LL = 1.5 kN/m^2 for top floor.

(longer)
 Ly

(shorter)
 Lx



one way slab: $\frac{Ly}{Lx} > 2$

Two way slab: $\frac{Ly}{Lx} \leq 2$

one-way slab

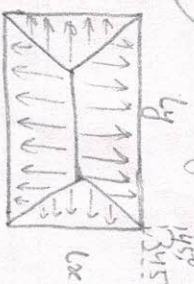


TL = Total load = (DL + LL) kN/m^2

load per m of longer slab = $TL \times \frac{Lx}{2}$

Case 1

Two way slab



IS-456:2000

24.5% loads on
supporting beams

TL = (DL + LL)
 kN/m^2

load per m of longer side = $TL \times \frac{Lx}{2} \left(Ly - \frac{Lx}{2} \right)$

$\frac{Ly}{4} = P_1$

load per m of shorter side = $TL \times \frac{Lx}{4} = P_2$

$$X_{Damp} = 0.05$$

$$\begin{matrix} M=1 & K_{init}=0 \\ K_{user}=0 & \text{switches} \\ K_{comm}=1 \end{matrix}$$

$$\omega_i = \sqrt{\lambda_i}$$

$$i=1$$

$$D = \alpha_M * M + \beta_{K_{user}} * K_{user} + \beta_{K_{init}} * K_{init}$$

$$+ \beta_{K_{comm}} * K_{comm}$$

$$\omega_j = \sqrt{\lambda_j}$$

$$j=3$$

$$\beta_{K_{user}} = K_{user} * \frac{2 * X_{Damp}}{\omega_i + \omega_j}$$

$$\beta_{K_{init}} = K_{init} * \frac{2 * X_{Damp}}{\omega_i + \omega_j}$$

$$\beta_{K_{comm}} = K_{comm} * \frac{2 * X_{Damp}}{\omega_i + \omega_j}$$

rayleigh lambda M, beta K, beta Kinit, beta Kcomm

Area, no. of bars, height of columns @ Circular

$$D_{CP} = \frac{200}{2} = 100$$

$$205:1994 \Rightarrow 25\%$$

$$905:2012 \Rightarrow 10\%$$

$$105:2020 \Rightarrow 25\%$$

$$205:2024 \Rightarrow 15\%$$

Return period earthquake.

$$D_{CP}: 456 = 114 * 4$$

$$205:1994: 992 = 124 * 8 = 496 + 496$$

$$905:2012: 496 = 124 * 4$$

$$105:2020: 528 = 264 * 2$$

$$205:2024: 496 = 124 * 4$$

Table 1: overall topologies

Table 2: Restraining

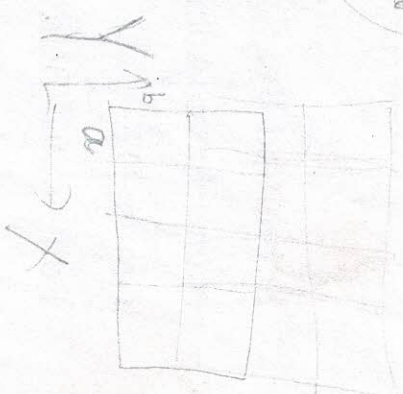
Table 3: Variation

Table 4: Reinforcement

Dataset generation
processing approach

$$\frac{n_x * n_y}{b} = \text{Aspect ratio}$$

$$a = \frac{\text{Aspect ratio} * \text{Area}}{n_x * n_y}$$



NBC 905:2024

Col: 8-3, 9-3

Beam: 8-3, 8-4, 9-4, 9-5

105set to 2003

$$\text{height} = \frac{2.85}{2.75}, 3.0, 3.55$$

$$2.75, 3.0, 3.55$$

$$5\% \text{ to } 10.5\%$$

$$K = \frac{f'_{cc}}{f'_{co}} = -1.254 + 2.254 \sqrt{1 + 7.94 \times \frac{f'_{co}}{f'_{eff}}}$$

$$- 2 \frac{f'_{eff}}{f'_{co}} \quad \text{--- ①}$$

① $f'_{co} = 20 \text{ MPa}$,

~~EPSCC~~ $EPSCC = EPSCO \times \left(1 + 5 \times \left(\frac{f'_{cc}}{f'_{co}} - 1 \right) \right)$

$f_{yk} = 250 \text{ MPa}$ (Stirrups yield stress)

$B, H = \text{dim. of column}$

Concrete cover = c

Core dimensions: $b_c, h_c = B - 2c, H - 2c$

② (ϕ_s) dia, spacing of stirrups

No. of " (2 legs in direction of H & B axes)

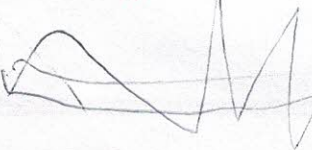
$$A_{sx} = \frac{\pi \phi_s^2}{4} \times 2$$

$$A_{sy} = \frac{\pi \phi_s^2}{4} \times 2$$

	CCP
NBC 205:1994	
205:2012	
105:2010	
205:2024	

$$T_m = 2\pi$$

$$\frac{\sum (C_i^2 \omega_i)}{\sum C_i^2}$$



③

$$s_x = \frac{A_{sx}}{s_{hc}}, \quad s_y = \frac{A_{sy}}{s_{bc}}$$

④

$$K_e = \left(1 + \frac{s'}{2b_c} \right) \left(1 - \frac{s'}{2h_c} \right), \quad s' = s - \phi_s$$

⑤

$$f'_{lex} = K_e s_x f_{yk}$$

$$f'_{ley} = K_e s_y f_{yk}$$

⑥

$$f'_{leg} = \min(f'_{lex}, f'_{ley}) \text{ or } \left(\frac{f'_{lex} + f'_{ley}}{2} \right)$$

Common (congruent) con- brg
w/ other (congruent)

Pgv / Pga ratio

Predominant Period (T_p)

Shear-wave Velocity

V_{s30}, T_p (PP), $A_I, P_{gA}, P_{gV}, P_{gA}$,
Significant Duration, $CAV, T_m, \beta_{SA}(T_p)$

$$T_m = \frac{\sum (C_i^2 T_i)}{\sum C_i^2}$$

Quality
Reverse Faulting
Calc. range

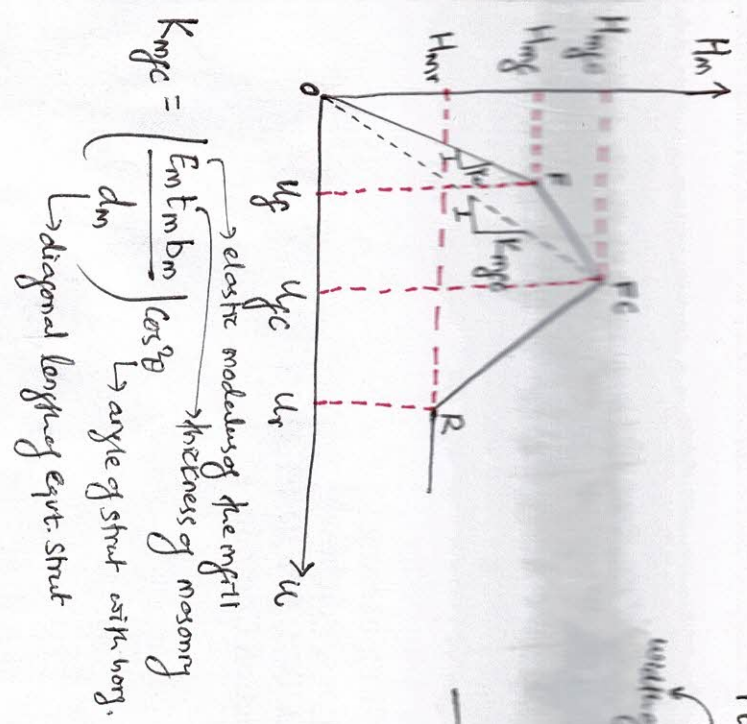
Concurrent paths

Org. Reinforcement ratio beam || column

designing Overstrength $[1.0, 0.0, 0.0]$
 $[0.1, 0.00]$

Normalization

FEMA 306,
 $\lambda_{bm} = 0.175 (\lambda_h)^{-0.4} d_m$
 (welding eqn. for strut)
 $\lambda_h = \left(\sqrt{\frac{E_m h_m d_m \cos \theta}{4 E_c I_m}} \right) h$



$$K_{mpe} = \left(\frac{E_m t_m b_m}{d_m} \right) \cos^2 \theta$$

elastic modulus of the masonry
 thickness of masonry
 angle of strut with horz.
 diagonal length of eqt. strut

OF = un-cracked phase

F-FC = post cracking phase

FE-Q = post peak shear strength deterioration

$$H_{mpe} = (H_{min}) K_m b_m \cos \theta$$

$$K_0 = 3 K_1 + 4 K_{mpe}, K_1 = \frac{1 + 6 \phi}{2 + 3 \phi} \left(\frac{42 E_c I_p}{h^3} \right)$$

$$K_2 = 0.02 K_{mpe}$$

$$H_{mpe} = 0.8 H_{mpe}$$

$$H_{mcr} = 0.35 H_{mpe}$$

$$\phi = \frac{I_{beam}}{I_{column}} \left(\frac{h}{l} \right)$$

Category case:

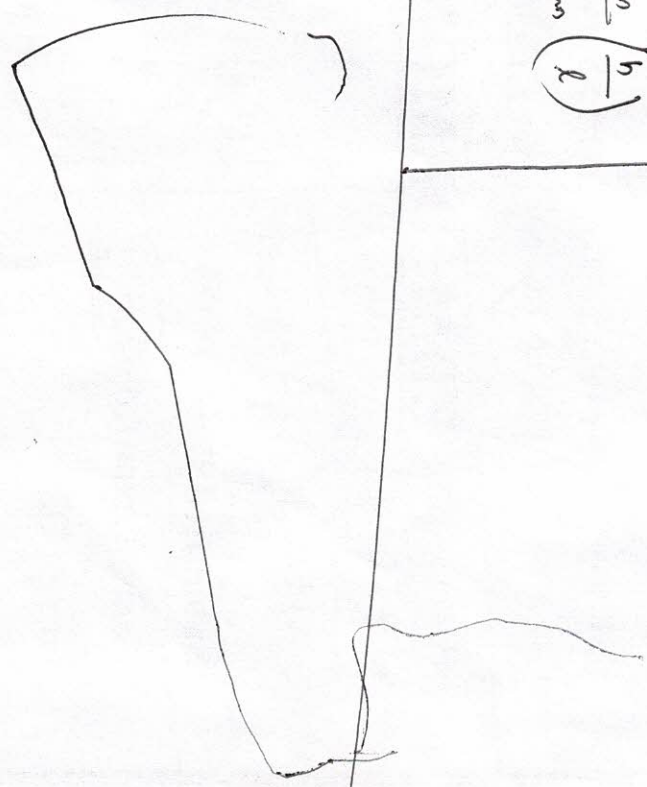
Structural: Beam Area NBC 405:2012, NBC 105:2002, NBC 205:2024

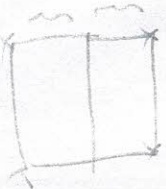
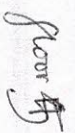
Bay width X, Y, Z : 3
 No of bays X, Y, Z : 3
 Aspect ratio : 1

Time period $T_1, T_2 : 2$

Seismic : T mean, PGA, PGV, PGV/PGA, AT, CAU, PP, SD, SA (T_L)

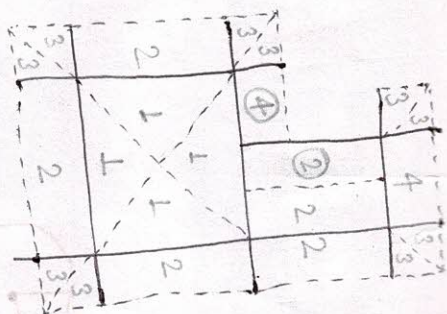
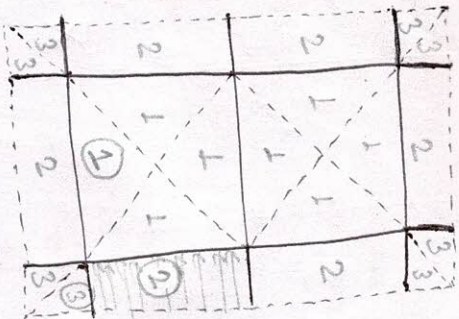
Dmg : 1st, 2nd, 3rd, 4th // Rest Any DI





3-16-30





wall load = $115 \times 10 \text{ ft} \times 120 \text{ mm}$
 on all beams



$$P_{11} = \frac{1}{2} \times 12 \text{ ft} \times 6 \text{ ft} \times T_L$$

$$P_{12} = 12 \text{ ft} \times 3 \text{ ft} \times T_L$$

$$P_{13} = \frac{1}{2} \times 3 \text{ ft} \times 3 \text{ ft} \times T_L$$

$$P_{14} = 6 \text{ ft} \times 3 \text{ ft} \times T_L$$

$T_L = \text{slab} + \text{floor finish} + \text{live load}$

$$P_{11} = P_{12} / P_{13} = 4.15$$

$$P_{12} = P_{13} / P_{14} = 4.15$$

$$P_{13} = P_{14} / 8 \text{ ft} = 2.07$$

$$P_{14} = P_{15} / 6 \text{ ft} = 4.15$$

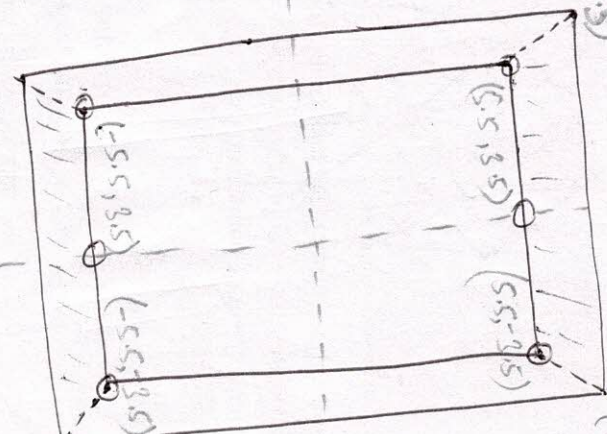
$$P_{11} = 1.887$$

$$P_{12} = 1.906$$

$$\text{wall load} = 7.01$$

(6.5, 4.5)

(-6.5, 4.5)

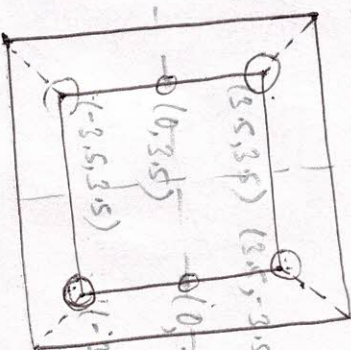


(6.5, -4.5)

(4.5, 4.5)

(4.5, -4.5)

(-4.5, 4.5)



XJ62-8WGL-YD4W-72RZ-MTYN-H7HD

452, 3.06
290, 3.06

153, 2.98
453, 2.98
292, 3.06
166, 3.48

2418

$$\frac{1.1908}{4.2808} = 1.0512$$



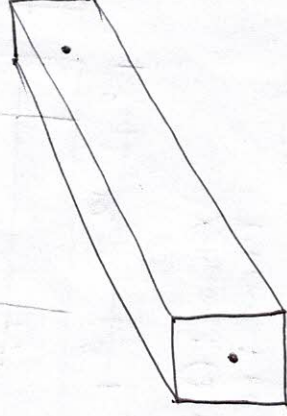
Tribhuban University
Institute of Engineering
Central Campus, Pulchowk
DEPARTMENT OF CIVIL ENGINEERING
Survey Instruction Committee

AS - $\frac{I 9}{65 (a-1)}$

BS - $\frac{9810}{45}$

- $\frac{9810 * 65}{I 9}$

= $\frac{9810 * 65 (rev)}{pg 2 (normal)}$



Report On
Surveying Practical

Course Surveying -1

CE-504

THEODOLITE & TOTAL STATION HANDLING & TRAVERSING

Submitted By:

Name : Laxmi Adhikari

Group: 

Roll No: 077BCE077

Submitted To:

Class Teacher:

Survey Instruction Committee

Department Of Civil Engineering

Pulchowk, Campus